

Exam 2

CH368 – Biophysics

Spring 2026

Show all your work and write all your explanations in full sentences.

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Honor Code: The core values of The University of Texas at Austin are learning, discovery, freedom, leadership, individual opportunity, and responsibility. Each member of the university is expected to uphold these values through integrity, honesty, trust, fairness, and respect toward peers and community.

SCORE: _____ / 150

1. (20 points) **Fourier Transforms.** Describe whether the following functions $F(\omega)$ will be real, imaginary, or mixed. Circle your answer and explain briefly in 1-2 sentences.

A. Real | **Imag** | Mixed:

$$F(\omega) = \int_{-\infty}^{\infty} t^3 e^{-i\omega t} dt$$

Since t^3 is odd and the Fourier transform of a real odd function is purely imaginary, the real part of the integrand is odd and cancels out over symmetric limits, leaving only the imaginary term.

B. **Real** | Imag | Mixed:

$$F(\omega) = \int_{-\infty}^{\infty} \cos^2(t) e^{-i\omega t} dt$$

Since $\cos^2(t)$ is even and the Fourier transform of a real even function is purely real, the imaginary part of the integrand is odd and cancels out over symmetric limits, leaving only the real even term.

2. (20 points) **Crowding.** Explain why studying globular proteins in dilute buffer solutions does not recapitulate the behavior of the same proteins in the cytosol? What are the two main components that describe the effects of crowding on protein thermodynamics?

In the cytosol, there are constant interactions with other proteins due to the crowded environment, whereas in a dilute buffer, interactions are far fewer. This crowded environment promotes fully folded protein states to occupy less volume than that of an unfolded protein. It also can impact the conformation of the protein through electrostatic and Van der Waal's interactions with neighboring proteins in the cell.

3. **Hydrophobicity and entropy (10 pts)** Consider a purely hydrophobic sequence of a certain length (HHHHH..... where H=hydrophobic residue), compared to a normal protein of the same length with a fraction of H and P (P=polar residue). Based on the hydrophobicity arguments we made in lecture would you expect the folding entropy ($\Delta S_{\text{folding}}$) of the (HHHHH.....) protein be larger/smaller/same as a protein of the same length but with H and P residues? Explain your thought process.

A purely hydrophobic sequence would have a larger $\Delta S_{\text{folding}}$. In the unfolded state, hydrophobic residues force surrounding water molecules to become ordered. When the protein folds and buries these residues, the ordered water molecules are released into bulk solvent, increasing entropy. Because a purely hydrophobic sequence buries more hydrophobic surface area than a mixed H/P protein, it results in a larger entropy increase upon folding.

4. (20 points) Hydrophobicity Scales: In class, we discussed several different hydrophobicity scales, there are 98 different scales in the literature.

a. (10 pts) What is the fundamental reason why there are multiple different scales?

Hydrophobicity is not a quantitative, measurable property of molecules, so scales are dependent on different properties, molecular interactions, and environments to describe hydrophobicity.

b. (10 pts) Describe, in detail, what experiments can be performed to construct a hydrophobicity scale?

There are many ways to construct a hydrophobicity scale. One method is an energy transfer scale, where the free energy change is measured when moving a side chain from the protein core to solution. Another common method is measuring the partitioning of amino acid side-chain analogs between water and an organic solvent (such as octanol) and calculating hydrophobicity from the partition coefficient.

5. (20 points) Hydrophobicity, entropy, and folding. Proteins will fold into a defined 3D structure if two conditions are met. First, the free energy of burying residues must be such that there is an energetic advantage to folding versus a random chain, where all residues are exposed to the solvent. What is the second condition that must be met? (hint: consider why there are only specific types of secondary structures in proteins).

After the burying of hydrophobic provides the thermodynamic driving force of protein folding, the amide and carbonyl groups on the amino acid backbones have hydrogen bonds that must be satisfied after they lost their interactions with water after folding. The need to satisfy their hydrogen bonding potential leads to the interactions that allow for low-energy β -sheet and α -helix geometries to form.

6. (20 points) Simulations. Consider this scenario: You are developing a new force field for molecular dynamics simulations and would like to benchmark this force field against experimental observables such as density. You find that when you compute the density of water from your simulation, the density is too high (compared to experiment). Provide a reasonable rationale for

how you may modify the force field to better match the experimental density. What are specific parameters that may be optimized?

The interaction energy between water molecules may be too high. If the hydrogen bonding interactions between water molecules are more favorable than in reality, the molecules will pack too closely together and increase the density of water in the simulation. Specific parameters that could be adjusted include the partial charges, which determine the strength of electrostatic interactions and hydrogen bonding, and the bond angle, which affects the molecular dipole and hydrogen bonding geometry. Additionally, adjusting the van der Waals parameters, such as the effective size σ , could increase the repulsive distance between molecules and bring the simulated density closer to the experimental value.

7. (20 points) Water as a biomolecule. In class we learned that simulations of hydrophobic “spheres” in water are a good model understanding how non-polar molecules interact with water. The simulations show that at a radius below ~ 1 nm the thermodynamic cost is entropic and above ~ 1 nm, the contribution becomes enthalpic. What is the significance of the ~ 1 nm length scale in water?

The 1nm length scale in water corresponds to the size of the hydrogen bond network, dictating how water organizes around hydrophobic surface. With molecules < 1 nm, water can readily form a solvation shell using hydrogen bonds around the molecule, lowering the entropy of the surrounding water molecules. With molecules > 1 nm, this network is unable to form as it would require the breaking of hydrogen bonds. At this length scale, the interaction is dominated by the enthalpic cost rather than the entropic ordering of water.

8. (10 points) Transitions in water. What is a “dewetting” transition and why is it significant in biology?

A dewetting transition is the sudden expulsion of water from a confined hydrophobic space, resulting in the formation of a transient “dry” region. This occurs when water molecules cannot form favorable interactions with hydrophobic surfaces in narrow cavities. In biology, this can be used to regulate ion transport through hydrophobic gates in ion channels and facilitate hydrophobic collapse during protein folding or promote protein aggregation

9. (10 points)_Fourier transforms. Consider this scenario. You are looking to sign up for an online music service such as Spotify. The free version includes audio quality sampled at 44.1 kHz (CD Quality), but the premium version comes with audio quality sampled at 192 kHz (“HD” Quality). Given that the human hearing range is 30 Hz to 20 kHz, would you be able to perceive an improvement in sound quality by paying for the premium subscription? Explain your reasoning using what you learned about Fourier-Transforms.

Using the Nyquist-Shannon sampling theorem, a continuous signal that can be heard by a human ear must have a sampling rate at least twice its highest frequency component (40 kHz in this scenario). If this minimum is satisfied, all the Fourier components in the audible range will be represented. Because both the free and premium versions satisfy this requirement, the average listener will not hear a difference between the two sampling rates.